

An adjoint proof of an ABP-like inequality on the Heisenberg group

Abstract

We obtain an ABP-like inequality for arbitrary uniformly elliptic horizontal coefficient matrices on the Heisenberg group. The proof uses adjoint power entropies along an auxiliary regularization. A central convolution allows scalar and matrix fields to evolve differently while preserving their double-divergence relation. Its matrix error is small enough to be absorbed. No commutation condition on the coefficient matrix, or smallness of the ellipticity ratio, is required.

1 Our goal

Use coordinates $\xi = (x, y, s)$ on $\mathbb{H}^n$, and write $z = (x, y)$,

$$X_k = \partial_{x_k} - \frac{y_k}{2} \partial_s, \quad Y_k = \partial_{y_k} + \frac{x_k}{2} \partial_s, \quad T = \partial_s.$$

Set $Z = (X_1, \dots, X_n, Y_1, \dots, Y_n)$, $m = 2n$, $Q = 2n + 2$, and

$$J = \begin{pmatrix} 0 & I_n \\ -I_n & 0 \end{pmatrix}, \quad \Delta_H = \sum_i Z_i^2, \quad \mathcal{D}B = \sum_{i,j} Z_i Z_j B_{ij}.$$

Thus $[Z_i, Z_j] = J_{ij}T$. All these fields are divergence-free. Matrix norms are Frobenius norms. The group law, dilations and balls are

$$(z, s) \circ (z', s') = (z + z', s + s' + \frac{1}{2}z^\top J z'), \quad \delta_R(z, s) = (Rz, R^2 s),$$

$$B_R = \{|z|^4 + 16s^2 < R^4\}, \quad B_R(\xi_0) = \xi_0 \circ B_R.$$

Consider $\mathcal{L}_A u = a_{ij} Z_i Z_j u$, where

$$A = A^\top, \quad \lambda I \leq A \leq \Lambda I, \quad 0 < \lambda \leq \Lambda < \infty. \quad (1)$$

For symmetric A one may equivalently use the symmetrized horizontal Hessian.

Theorem 1.1. *For every $n \geq 1$ and $0 < \lambda \leq \Lambda$, there exist $1 < p < \infty$ and $C < \infty$, depending only on n, λ, Λ , with the following property. Let $R > 0$, $\Omega \subset B_R(\xi_0)$ be bounded and open, and let A be measurable and satisfy (1) almost everywhere. If $u \in C^2(\Omega) \cap C(\overline{\Omega})$ and $f \in C(\Omega)$ satisfy*

$$-\mathcal{L}_A u \leq f \quad \text{a.e. in } \Omega, \quad u \leq 0 \quad \text{on } \partial\Omega,$$

then

$$\|u_+\|_{L^\infty(\Omega)} \leq C R^{2-Q/p} \left(\int_{\Omega \cap \{u>0\}} f_+^p d\xi \right)^{1/p}. \quad (2)$$

The assertion is understood when the right-hand side is finite. No modulus of continuity of A , and no regularity of $\partial\Omega$, enters the constants.

Applying the theorem to u and $-u$ gives

$$\|u\|_\infty \leq C R^{2-Q/p} \|f\|_{L^p(\Omega)}$$

for $-\mathcal{L}_A u = f$ with zero boundary data. We do not seek an optimal exponent.

The proof uses two power entropies, as in the elliptic case: one for the adjoint density and one for the regularized coefficients. In the elliptic case, ordinary heat flow commutes with double divergence. On the Heisenberg group this commutation fails, so we construct scalar and tensor regularizations that together preserve the adjoint equation. We estimate the additional entropy and coefficient errors below and absorb them by choosing the power sufficiently close to one.

2 The adjoint equation

First suppose that $\Omega \subset B_1$ is smooth and A is smooth near $\overline{\Omega}$. Introduce

$$\mathcal{L}_{A,\varepsilon} = \mathcal{L}_A + \varepsilon T^2, \quad \varepsilon > 0.$$

This operator is uniformly elliptic in Euclidean coordinates on the bounded domain. It is enough to prove, for some $1 < r \leq 4/3$,

$$\|v\|_{L^r(\Omega)} \leq C\|g\|_{L^1(\Omega)}, \quad -\mathcal{D}(Av) - \varepsilon T^2 v = g, \quad v|_{\partial\Omega} = 0, \quad (3)$$

uniformly in ε , in derivatives of A , and in $\Omega \subset B_1$. By positivity and homogeneity, take $g \geq 0$, $\int g = 1$, and $v \geq 0$. Smooth data suffice. Testing with

$$\Phi(z, s) = \frac{1 - |z|^2}{2m\lambda}, \quad -\mathcal{L}_{A,\varepsilon}\Phi = \frac{\text{tr } A}{m\lambda} \geq 1,$$

gives

$$M := \int_{\Omega} v \, d\xi \leq \frac{1}{2m\lambda}. \quad (4)$$

Indeed, the boundary term is $\Phi\beta$, where

$$\beta = (\nu_H \cdot A\nu_H + \varepsilon\nu_T^2)\partial_\nu v \leq 0, \quad \nu_H = (Z_i \cdot \nu)_i, \quad \nu_T = T \cdot \nu.$$

Extend v, g by zero and extend A preserving (1). The same computation gives

$$-\mathcal{D}(Av) - \varepsilon T^2 v = g + \beta \sigma_{\partial\Omega} \leq g \quad \text{on } \mathbb{H}^n. \quad (5)$$

All unspecified integrals below are over $\mathbb{H}^n$. Define

$$|DF|_\varepsilon^2 = \sum_i |Z_i F|^2 + \frac{\varepsilon}{\lambda} |TF|^2, \quad \mathcal{H}_\varepsilon = \Delta_H + \frac{\varepsilon}{\lambda} T^2, \quad H_h = e^{h\mathcal{H}_\varepsilon}.$$

For matrices the gradient norm is summed over all components.

3 Scalar and tensor regularization

Let κ be a large constant, to be chosen later, and let C_h denote central convolution with

$$c_h(s) = \frac{1}{2\kappa h} \text{sech} \frac{\pi s}{2\kappa h}, \quad \widehat{c}_h(\tau) = \text{sech}(\kappa h \tau). \quad (6)$$

Write

$$P_h = H_h C_h, \quad \mathcal{K}B = JB - BJ, \quad S_h = H_h C_h e^{2hT\mathcal{K}}. \quad (7)$$

The last formula denotes the combined central multiplier; the isolated exponential need not define a bounded operator.

Lemma 3.1. *For sufficiently large κ , the tensor operator S_h is well defined, preserves symmetric matrix fields, and satisfies*

$$\mathcal{D}S_h B = P_h \mathcal{D}B, \quad |S_h B - P_h B| \leq \frac{C_n}{\kappa} P_h |B|. \quad (8)$$

The scalar operator P_h preserves positivity and mass. Moreover,

$$\|P_h F\|_\infty \leq C_n h^{-Q/2} \|F\|_1, \quad |T \log P_h F| \leq \frac{\pi}{2\kappa h} \quad (F \geq 0, F \not\equiv 0). \quad (9)$$

All constants are independent of h, ε, κ above a fixed threshold.

Proof. The commutation relations give

$$[\Delta_H, Z_i] = -2T J_{ik} Z_k, \quad \mathcal{H}_\varepsilon \mathcal{D}B = \mathcal{D}(\mathcal{H}_\varepsilon B + 2TKB).$$

Exponentiating proves the first identity in (8). To justify the operator, $\mathcal{K}$ is skew-adjoint on symmetric matrices and $\|\mathcal{K}\| \leq 2$. The central kernel of $C_h e^{2hT\mathcal{K}}$ is

$$M_h(s) = \frac{1}{2\kappa h} \left[\cosh \left(\frac{\pi s}{2\kappa h} I + \frac{\pi}{\kappa} \mathcal{K} \right) \right]^{-1}.$$

On an eigenvector with eigenvalue $i\beta$, this is $c_h(s + 2ih\beta)$. For real t and small θ ,

$$|\cosh(t + i\theta)| \geq \cos \theta \cosh t, \quad \left| \frac{1}{\cosh(t + i\theta)} - \frac{1}{\cosh t} \right| \leq C|\theta| \operatorname{sech} t.$$

Consequently

$$\|M_h(s) - c_h(s)I\| \leq \frac{C_n}{\kappa} c_h(s), \quad \int M_h(s) ds = I. \quad (10)$$

The mass follows also by setting $\tau = 0$ in the Fourier multiplier. This proves the pointwise estimate in (8). For the exponentiation, first impose compact central Fourier support and then remove the Fourier cutoff using the integrable combined kernel. This also proves the identities for compactly supported distributions.

The heat bound $\|H_h\|_{1 \rightarrow \infty} \leq C_n h^{-Q/2}$ is uniform in ε , because the additional central heat flow is an L^1 contraction. Central convolution is also an L^1 contraction. Finally, $|c'_h| \leq \pi c_h/(2\kappa h)$ proves the logarithmic derivative bound. $\square$

Remark 3.2. For completeness, the only heat-kernel input in the proof is the preceding dispersion bound. With Fourier transform in s , the kernel of $e^{h\Delta_H}$ has transform

$$\widehat{k}_h(z, \tau) = (4\pi h)^{-n} \left(\frac{h\tau}{\sinh(h\tau)} \right)^n \exp \left(-\frac{|z|^2}{4h} h\tau \coth(h\tau) \right).$$

It follows by the radial Gaussian ansatz for the Fourier-transformed heat equation. Fourier inversion and integrability of $(t/\sinh t)^n$ give $\|k_h\|_\infty \leq C_n h^{-n-1}$. Positivity and unit mass are the usual heat-flow properties. See also [1, Section 2, equation (2.7)].

Set

$$w_h = P_h v, \quad g_h = P_h g, \quad B_h = S_h(Av), \quad a_h = B_h/w_h. \quad (11)$$

For $\kappa \geq \kappa_0(n, \lambda, \Lambda)$, the pointwise tensor error and positivity imply

$$\frac{\lambda}{2} I \leq a_h \leq 2\Lambda I. \quad (12)$$

Indeed, $\lambda w_h I \leq P_h(Av) \leq \Lambda w_h I$ and $P_h|Av| \leq \sqrt{m}\Lambda w_h$. Applying P_h to (5) gives the exact inequality

$$-\mathcal{D}(a_h w_h) - \varepsilon T^2 w_h \leq g_h. \quad (13)$$

No derivatives of the original coefficients have been estimated.

4 The scalar power entropy

Fix $1 < r \leq 4/3$, put $\chi = r(r-1)$, and write

$$E(h) = \int w_h^r d\xi, \quad I(h) = \chi \int w_h^{r-2} |Dw_h|_\varepsilon^2 d\xi, \quad \mathcal{I} = \int_0^1 I(h) dh, \quad E_0 = \int v^r d\xi.$$

At this stage v is bounded and compactly supported, so $E_0 < \infty$. Markov Jensen gives $E(h) \leq E_0$ before any entropy estimate. We work on $0 < h < 1$, so a cutoff of the power entropy at small values is unnecessary.

Lemma 4.1. *There is C_n , independent of r, ε, κ , such that*

$$\mathcal{I} \leq E_0, \quad E_0 \leq E(1) + (1 + C_n \kappa) \mathcal{I}. \quad (14)$$

Proof. The logarithmic derivative of $\widehat{c}_h$ is $-\kappa\tau \tanh(\kappa h\tau)$. It is the symbol of a symmetric jump generator with density

$$N_h(t) = \frac{\pi \cosh \theta}{4\kappa h^2 \sinh^2 \theta}, \quad \theta = \frac{\pi|t|}{2\kappa h}. \quad (15)$$

One obtains this by differentiating

$$\log \cosh(a\tau) = \int_{\mathbb{R}} \frac{1 - \cos(\tau t)}{2|t| \sinh(\pi|t|/(2a))} dt.$$

This identity follows, in turn, by the product expansion of $\cosh$ and the elementary integral for $\log(1 + \tau^2/b^2)$. Thus

$$-E'(h) = I(h) + J(h), \quad J(h) \geq 0,$$

where, writing $w = w_h$ and $\xi_t = \xi \circ (0, t)$,

$$J(h) = \frac{r}{2} \int_{\mathbb{R}} \int_{\mathbb{R}} N_h(t) (w(\xi_t) - w(\xi)) (w(\xi_t)^{r-1} - w(\xi)^{r-1}) dt d\xi.$$

For $a, b > 0$ and $1 < r \leq 4/3$,

$$(a - b)(a^{r-1} - b^{r-1}) \leq C(r-1)(1 + |\log(a/b)|)(a^{r/2} - b^{r/2})^2. \quad (16)$$

To see this, put $a/b = e^\ell \geq 1$. For $\ell \leq 1$ use the mean value theorem. For $\ell \geq 1$, use $e^{(r-1)\ell} - 1 \leq (r-1)\ell e^{(r-1)\ell}$ and $e^{r\ell/2} - 1 \geq (1 - e^{-1/2})e^{r\ell/2}$.

By (9), $|\log(w(\xi_t)/w(\xi))| \leq \theta$. Also $t^2 N_h(t)(1 + \theta) \leq C\kappa$. With $q = w^{r/2}$, the fractional-energy formula therefore gives

$$J(h) \leq C\kappa\chi \langle q, |T|q \rangle \leq C_n \kappa \chi \int |\nabla_H q|^2 d\xi \leq C_n \kappa I(h).$$

Here $|T|$ has central Fourier symbol $|\tau|$, and

$$n \langle q, |T|q \rangle \leq \int |\nabla_H q|^2 d\xi. \quad (17)$$

Indeed, after central Fourier transform, $[X_k, Y_k] = i\tau$ and integration by parts gives $|\tau| \|q\|_2^2 \leq \|X_k q\|_2^2 + \|Y_k q\|_2^2$. Sum in k and integrate in τ . Integrating the entropy identity proves (14). $\square$

5 The coefficient power entropy

The point of the tensor correction is that its derivative error is quadratic in κ^{-1} .

Lemma 5.1. *For $1 < r \leq 4/3$ and $\kappa \geq \kappa_0$, there is $C = C(n, \Lambda)$, independent of r, ε, κ , such that*

$$\chi \int_0^1 \int w_h^r |Da_h|_\varepsilon^2 d\xi dh \leq C(\chi + \kappa^{-2}) E_0 + C\kappa^{-2} \mathcal{I}. \quad (18)$$

Proof. Write

$$q_h = H_h v, \quad R_h = H_h(Av), \quad b_h = R_h/q_h, \quad \bar{a}_h = P_h(Av)/w_h.$$

For two functions q, qb evolving by $\mathcal{H}_\varepsilon$, the quotient computation gives

$$\begin{aligned} (\partial_h - \mathcal{H}_\varepsilon)(q^r |b|^2) &= -r(r-1)q^{r-2}|b|^2 |Dq|_\varepsilon^2 - 2q^r |Db|_\varepsilon^2 \\ &\quad - 4(r-1)q^{r-1} \sum_\alpha b_\alpha \langle Dq, Db_\alpha \rangle_\varepsilon \\ &\leq -c_r q^r |Db|_\varepsilon^2, \quad c_r = \frac{2(2-r)}{r} \geq 1. \end{aligned}$$

Young's inequality gives the last line. The same integrated inequality holds for the common scalar flow P_h : the extra jump contribution is nonpositive because

$$(q, R) \mapsto q^{r-2} |R|^2$$

is convex for $1 \leq r \leq 2$. It follows that

$$\int_0^1 \int q_h^r |Db_h|_\varepsilon^2 d\xi dh + \int_0^1 \int w_h^r |D\bar{a}_h|_\varepsilon^2 d\xi dh \leq 2m\Lambda^2 E_0. \quad (19)$$

Let $\mathcal{N}_h$ be central convolution with $M_h - c_h I$, and put

$$F_h = \mathcal{N}_h R_h, \quad e_h = F_h/w_h, \quad a_h = \bar{a}_h + e_h.$$

The central convolution commutes with all the fields in D . From (10),

$$|e_h| \leq C\Lambda/\kappa, \quad \int w_h^{r-2} |DF_h|_\varepsilon^2 d\xi \leq C\kappa^{-2} \int q_h^{r-2} |DR_h|_\varepsilon^2 d\xi. \quad (20)$$

The second inequality is weighted Jensen. Explicitly, for a positive probability convolution C ,

$$(Cq)^{r-2} (C|F|)^2 \leq C(q^{r-2} |F|^2),$$

by Cauchy-Schwarz and $C(q^{2-r}) \leq (Cq)^{2-r}$. Apply this to C_h and use $|\mathcal{N}_h F| \leq C\kappa^{-1} C_h |F|$.

Since $DR_h = q_h Db_h + b_h Dq_h$ and $|b_h| \leq \sqrt{m}\Lambda$, the quotient rule in (20) gives

$$\begin{aligned} \int w_h^r |De_h|_\varepsilon^2 d\xi &\leq C\kappa^{-2} \int q_h^r |Db_h|_\varepsilon^2 d\xi \\ &\quad + C\kappa^{-2} \left(\int q_h^{r-2} |Dq_h|_\varepsilon^2 d\xi + \int w_h^{r-2} |Dw_h|_\varepsilon^2 d\xi \right). \end{aligned}$$

The ordinary heat power identity gives

$$\chi \int_0^1 \int q_h^{r-2} |Dq_h|_\varepsilon^2 d\xi dh \leq E_0.$$

Multiply the preceding error estimate by χ , integrate in h , and use (19). Together with $|Da_h|^2 \leq 2|D\bar{a}_h|^2 + 2|De_h|^2$, this proves (18). $\square$

6 Proof of the estimate

Test (13) with rw_h^{r-1} . One integration by parts gives

$$\chi \int w_h^{r-2} (a_h \nabla_H w_h \cdot \nabla_H w_h + \varepsilon |Tw_h|^2 + w_h \nabla_H w_h \cdot \operatorname{div}_H a_h) d\xi \leq r \int w_h^{r-1} g_h d\xi.$$

Here $(\operatorname{div}_H a)_i = Z_j a_{ij}$. By (12) and Young's inequality,

$$I(h) \leq \frac{4m}{\lambda^2} \chi \int w_h^r |Da_h|_\varepsilon^2 d\xi + \frac{4r}{\lambda} \int w_h^{r-1} g_h d\xi. \quad (21)$$

All computations may first be integrated over $\delta < h < 1$ with compact spatial cutoffs. For dilated cutoffs ζ_L , $|\nabla_H \zeta_L| = O(L^{-1})$, $|T\zeta_L| = O(L^{-2})$, and $\int t^2 N_h(t) dt = 2\kappa^2 h$. Thus the scalar and coefficient entropy cutoff errors vanish by Markov Jensen and $E_0 < \infty$. Their dissipation bounds then control the PDE flux cutoff errors by Cauchy–Schwarz. Send $L \rightarrow \infty$ at fixed ε, κ, r , and then $\delta \downarrow 0$. The limiting estimates have exactly the uniform constants displayed above.

By (4) and (9),

$$\|w_h\|_\infty \leq C(n, \lambda) h^{-Q/2}, \quad \int g_h d\xi = 1.$$

Choose also $r - 1 \leq 1/Q$. Then

$$E(1) \leq C(n, \lambda), \quad \int_0^1 \int r w_h^{r-1} g_h d\xi dh \leq C \int_0^1 h^{-Q(r-1)/2} dh \leq C(n, \lambda). \quad (22)$$

Combining (21) and Lemma 5.1, with constants uniform in r, ε, κ , gives

$$\mathcal{I} \leq C_0(\chi + \kappa^{-2})E_0 + C_0\kappa^{-2}\mathcal{I} + C_1. \quad (23)$$

Here and below C_0, C_1 depend only on n, λ, Λ . Choose $\kappa \geq \kappa_0$ so large that

$$C_0\kappa^{-2} \leq \frac{1}{2}, \quad 2C_0(1 + C_n\kappa)\kappa^{-2} \leq \frac{1}{4}.$$

With this κ fixed, choose $r > 1$ so close to one that

$$r \leq \frac{4}{3}, \quad r - 1 \leq 1/Q, \quad 2C_0(1 + C_n\kappa)r(r - 1) \leq \frac{1}{4}.$$

These choices depend only on n, λ, Λ . Lemma 4.1, (22), and (23) now give

$$E_0 \leq C + \frac{1}{2}E_0.$$

Thus $\int v^r \leq C$. Homogeneity and positivity prove (3) for general g , and duality gives the regularized maximum estimate with $p = r/(r - 1)$.

For the positive-part form, choose smooth convex φ_δ with $\varphi_\delta = 0$ on $(-\infty, 0]$, $0 \leq \varphi'_\delta \leq 1$, and $\varphi_\delta(t) \rightarrow t_+$. If $w \leq 0$ on the boundary and $-\mathcal{L}_{A,\varepsilon}w \leq F$, the classical chain rule gives

$$-\mathcal{L}_{A,\varepsilon}\varphi_\delta(w) \leq F_+ \mathbf{1}_{\{w>0\}}.$$

Both $\varphi_\delta(w)$ and the adjoint solution v vanish on the boundary. Hence

$$\int g \varphi_\delta(w) d\xi = \int v [-\mathcal{L}_{A,\varepsilon}\varphi_\delta(w)] d\xi \leq C \|F_+ \mathbf{1}_{\{w>0\}}\|_{L^p}.$$

Let $\delta \downarrow 0$ and take the supremum over nonnegative smooth g with $\int g = 1$. We obtain

$$\|w_+\|_\infty \leq C \|F_+ \mathbf{1}_{\{w>0\}}\|_{L^p}. \quad (24)$$

We finish the a priori passage for Theorem 1.1. At unit scale fix $a > 0$. The compact set $\{u \geq a\}$ lies in Ω . If nonempty, choose a smooth $D \Subset \Omega$ containing it, so $u - a < 0$ on ∂D . Extend A preserving its bounds and let A_j be ordinary Euclidean mollifications. Then $A_j \rightarrow A$ almost everywhere and

$$E_j := |(A_j - A)_{i\ell} Z_i Z_\ell u| \rightarrow 0 \quad \text{in } L^p(D),$$

because u is C^2 near $\overline{D}$. On D ,

$$-\mathcal{L}_{A_j,\varepsilon}(u - a) \leq f + E_j + \varepsilon |T^2 u|.$$

Apply (24) and let $j \rightarrow \infty$, then $\varepsilon \rightarrow 0$. Since D contains $\{u > a\}$,

$$\|(u - a)_+\|_{L^\infty(\Omega)} \leq C \|f_+ \mathbf{1}_{\{u>0\}}\|_{L^p(\Omega)}.$$

The inequality is trivial if the compact set is empty. Let $a \downarrow 0$. Finally, the change of variables $\xi = \xi_0 \circ \delta_R \zeta$ multiplies the equation by R^2 and volume by R^Q , proving (2).

Remark 6.1. If $AJ = JA$, the tensor correction disappears because $\mathcal{K}(Av) = 0$. For general A , applying the same scalar heat flow to v and Av is insufficient: already on $\mathbb{H}^1$,

$$[\Delta_H, X^2](sxy) = -4.$$

The proof above replaces exact componentwise commutation by exact scalar–tensor intertwining. The two estimates that permit this replacement are the $O(\kappa)$ scalar entropy cost and the $O(\kappa^{-2})$ coefficient energy error. After choosing κ , a power sufficiently close to one absorbs the remaining coefficient entropy.

References

- [1] N. Eldredge, *Gradient estimates for the subelliptic heat kernel on H-type groups*, J. Funct. Anal. **258** (2010), 504–533.
<https://arxiv.org/abs/0904.1781>.